

Dirichlet Convolution Presentation Notes

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1 Motivating Problem

$$\left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \left(\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}\right) = 1$$

Proof:

Let $f(n)$ be a multiplicative function such that

$$\left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \left(\sum_{n=1}^{\infty} \frac{f(n)}{n^s}\right) = 1$$

For $s > 1$, both series converge, so we can multiply them term by term, which gives

$$\sum_{a=1}^{\infty} \sum_{b=1}^{\infty} \frac{f(b)}{(ab)^s} = 1$$

Now we can group terms by $n = ab$, since every factorization of n , written $n = ab$, corresponds to choosing some b such that $b \mid n$. So, the series becomes,

$$\sum_{n=1}^{\infty} \frac{1}{n^s} \sum_{b \mid n} f(b) = 1$$

Because the series are equal, their coefficients must be equal for each n . If their product is equal to 1, or $\frac{1}{1^s}$, then

$$\sum_{b \mid n} f(b) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases}$$

Recall that this is the formula for the child of f . Since the definition of the child of f is the same as the definition of z , we can conclude that f is $\mu(n)$, the parent of z .

So we just showed that

$$\left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \left(\sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}\right) = \sum_{n=1}^{\infty} \frac{z(n)}{n^s}$$

Or, in other words

$$\sum_{d \mid n} \mu(d) m\left(\frac{n}{d}\right) = z(n)$$

2 Definition

This operation of a product on two series of this specific form can be defined as the following:

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} \sum_{n=1}^{\infty} \frac{g(n)}{n^s} = \sum_{n=1}^{\infty} \frac{(f * g)(n)}{n^s}$$

where

$$(f * g) = \sum_{d \mid n} f(d) g\left(\frac{n}{d}\right)$$

Or, alternatively,

$$(f * g) = \sum_{ab=n} f(a)g(b)$$

The second definition is particularly useful, since it is more symmetric than the first, and can be used to prove some important properties of Dirichlet convolution.

3 Commutativity

The first property we will talk about is commutativity. That is, $f * g = g * f$.

Start with our more symmetric definition of Dirichlet convolution:

$$f * g = \sum_{ab=n} f(a)g(b)$$

and

$$g * f = \sum_{ab=n} g(a)f(b)$$

Notice that these variables a and b are only defined in so far as their product is n . Therefore, we can just switch them

$$f * g = \sum_{ab=n} f(a)g(b) = \sum_{ba=n} f(b)g(a) = \sum_{ab=n} g(a)f(b) = g * f$$

4 Associativity

We can use a similar argument to prove associativity.

$$((f * g) * h) = \sum_{ab=n} (f * g)(a)h(b) = \sum_{ab=n} \left(\sum_{cd=a} f(c)g(d) \right) h(a) = \sum_{ab=n} \sum_{cd=a} f(c)g(d)h(a)$$

Notice that the conditions $ab = n$ and $cd = a$ implies that $bcd = n$. Regroup the terms as $c(db) = n$

$$\sum_{c(db)=n} f(c) \left(\sum_{db=m} g(d)h(b) \right) = \sum_{c(db)=n} f(c)(g * h)(m) = (f * (g * h))$$

5 Identity element

Now that we have discussed these properties, let me go back to $z(n)$. let n have divisors $1, d_1, d_2, \dots, n$, and let $f(n)$ be any multiplicative function. Then the convolution of $f * z$ is

$$\sum_{d|n} z(d)f\left(\frac{n}{d}\right) = z(1)f\left(\frac{n}{1}\right) + z(d_1)f\left(\frac{n}{d_1}\right) + \dots + z(n)f\left(\frac{n}{n}\right)$$

However, $z(n) = 0$ for $n > 1$, so every other term becomes 0 except for the first, giving

$$\sum_{d|n} z(d)f\left(\frac{n}{d}\right) = z(1)f(n) = f(n)$$

Therefore, $z(n)$ is the identity element under Dirichlet convolution.

6 Known Functions

Now, I will introduce a few interesting relationships between functions we have seen in this course that I came across using Dirichlet convolution:

$$m * m = \tau$$

$$\mu * \text{id} = \phi$$

$$\tau * \phi = \sigma$$

$$\sigma * \mu = \text{id}$$

$$\text{id} * m = \sigma$$

7 Open question

In this presentation so far, we have seen that Multiplicative functions under Dirichlet convolution follow commutativity, associativity, and have an identity element. To be a group, we would also need to prove the existence of inverses. That is, for every multiplicative function f , there exists a function f^{-1} such that

$$f * f^{-1} = z$$

There is one function that I know this doesn't work for, which I have used gamma to denote, and is defined like this

$$\gamma(n) := 0 \quad \forall n \in \mathbb{Z}^+$$

My goal was to prove that for every other function, inverses either must exist or don't always exist, but I was unable to do so before the due date of the presentation.